

NanoGT Match Report: Metachromatic leukodystrophy

Disease: Metachromatic leukodystrophy (ORPHA:512)

Primary gene: ARSA

Gene CDS: 1521 bp

Inheritance: Autosomal recessive

Target tissues: CNS

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	9.1/10	[GREEN] High	approved
2	Skysona	LV	8.4/10	[GREEN] High	approved
3	ABO-101	AAV9	8.3/10	[GREEN] High	phase1/2
4	RGX-121	AAV9	8.3/10	[GREEN] High	phase3
5	AVR-RD-01	LV	8.0/10	[GREEN] High	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 9.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
TOTAL (normalised)	9.11	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- Precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

Match #2: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 8.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.00	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
TOTAL (normalised)	8.39	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- Precedent target match: cns
- Partial match: extracellular/secreted component present
- LOF inheritance — compatible for gene replacement
- Pathway match: peroxisomal
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

Match #3: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
TOTAL (normalised)	8.28	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1521bp / cargo 4700bp (32% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected

- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

Match #4: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
TOTAL (normalised)	8.28	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1521bp / cargo 4700bp (32% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage

- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

Match #5: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
TOTAL (normalised)	8.00	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk